

Armenian Epigraphic OCR

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This research focuses on the detection and recognition of Armenian epigraphic inscriptions carved in stone, using a combination of classical computer vision and machine learning techniques. The goal is to develop tools for automating the documentation and analysis of Armenian cultural heritage.

1 Introduction

Armenian stone inscriptions, carved over centuries onto khachkars and monuments, represent a vital cultural and historical archive. Despite their significance, the field remains almost entirely unexplored, with only a single academic paper addressing the use of modern methods for their analysis^{[1][2]}. The deterioration of stone surfaces and the complexity of ancient scripts pose major challenges to traditional documentation efforts.

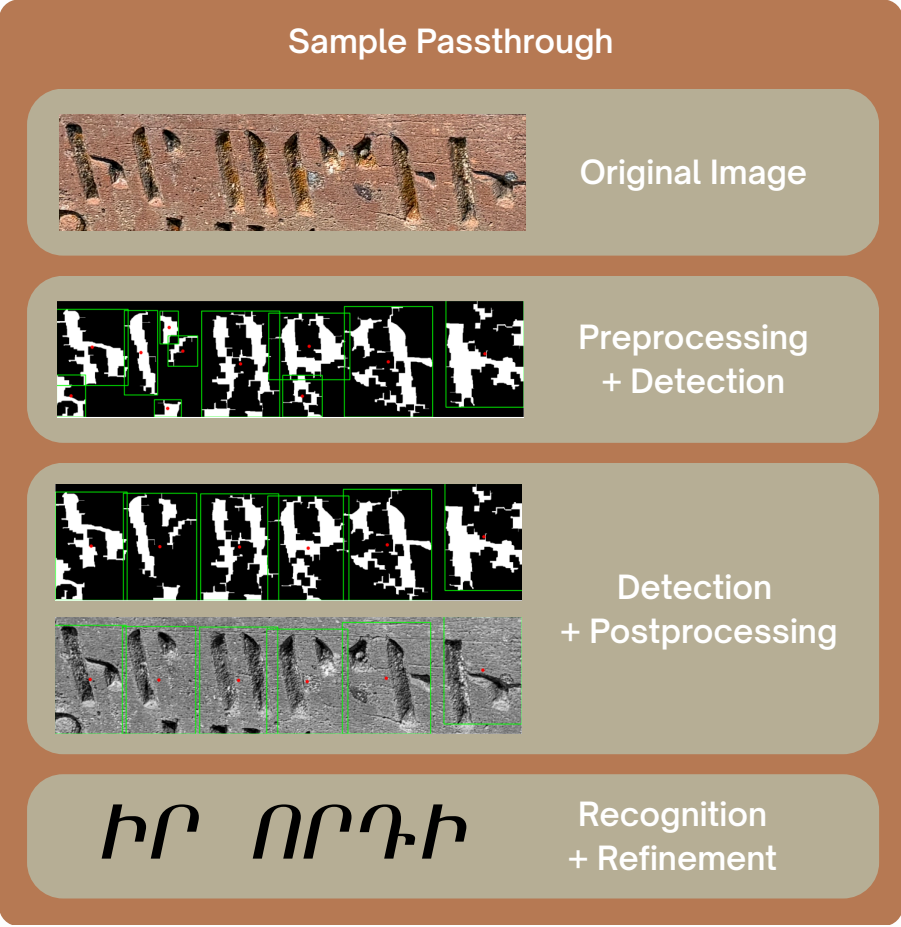
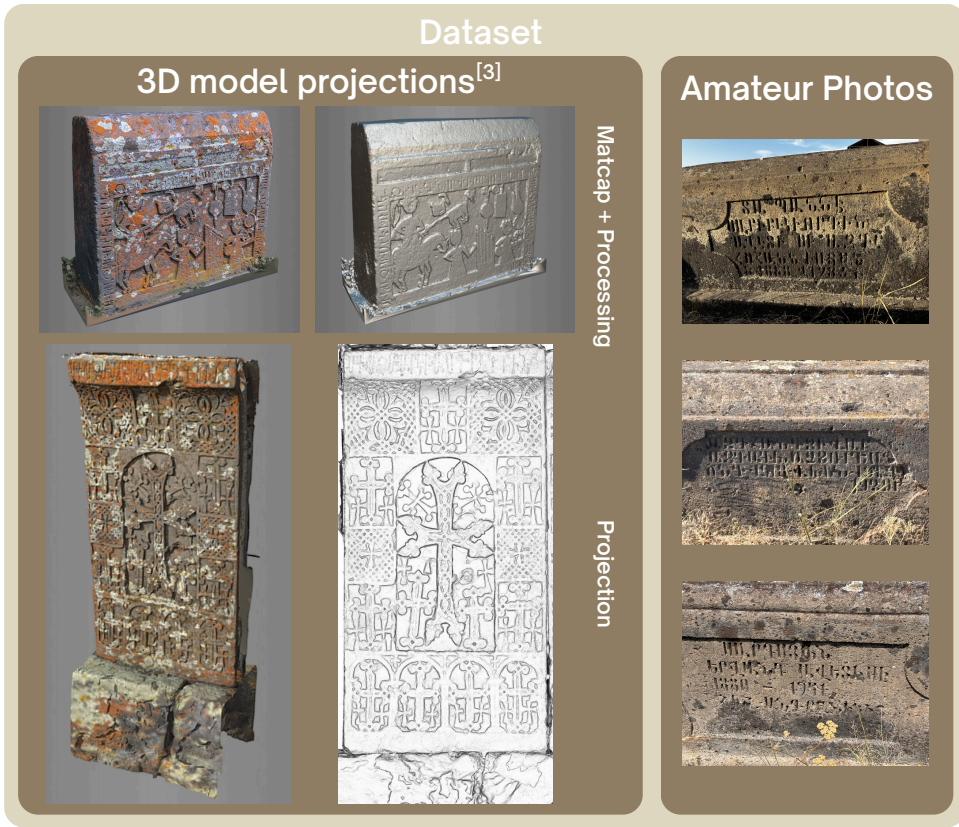
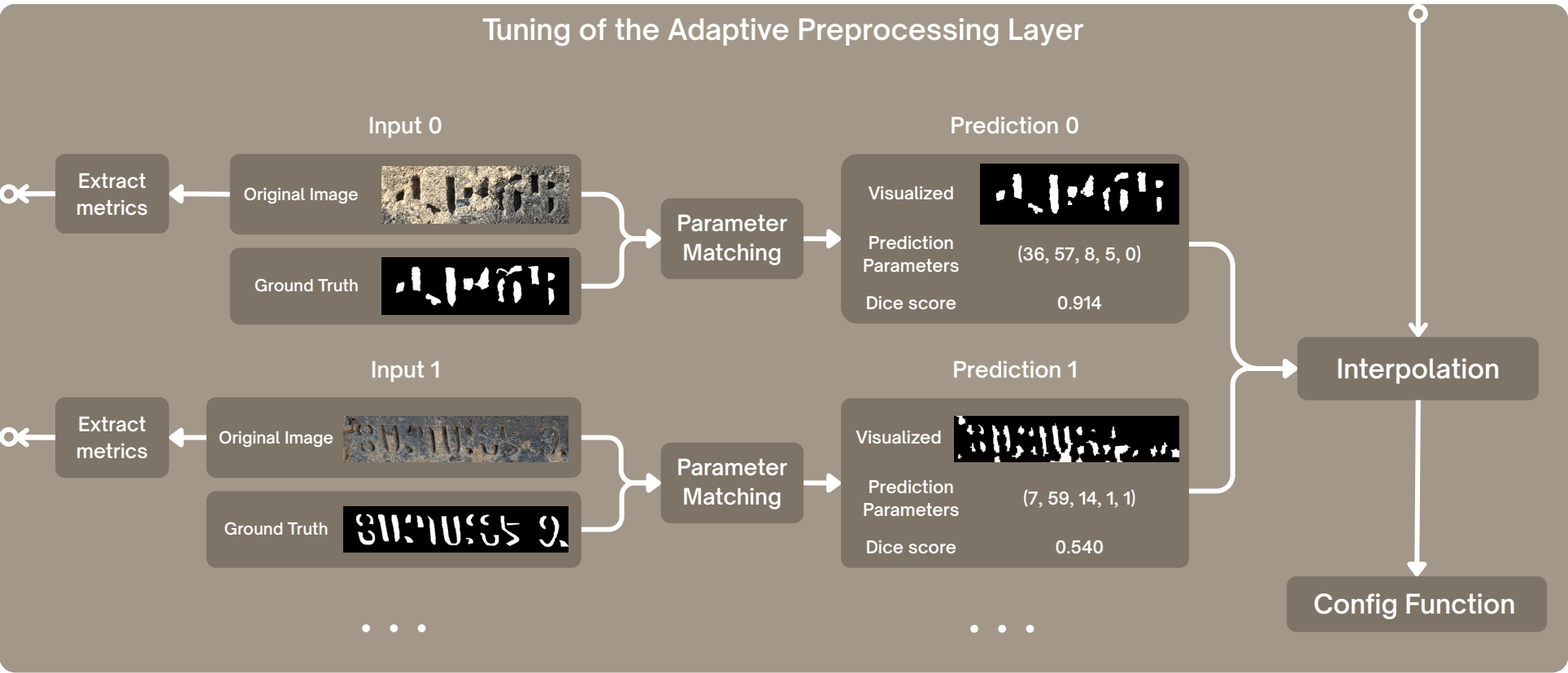
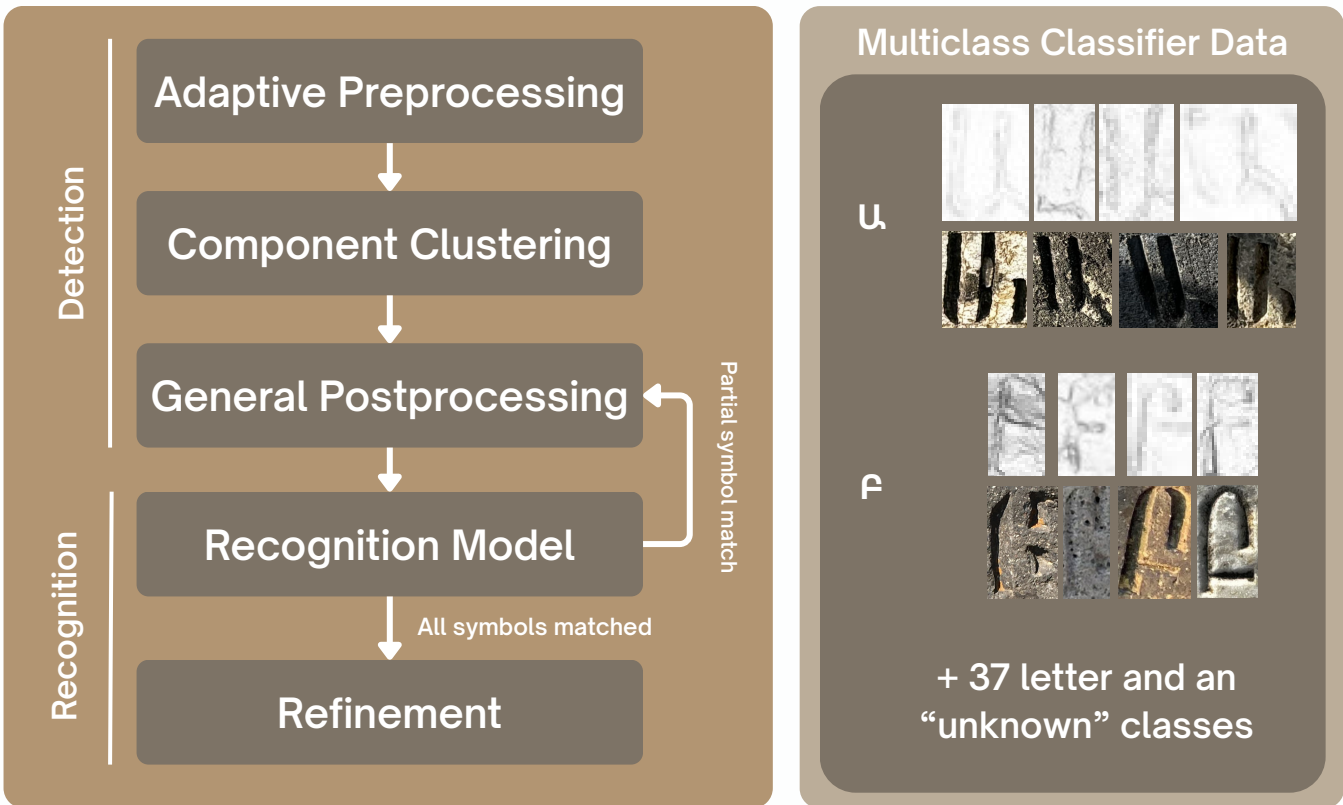
Problem: Manual transcription is slow, error-prone, and often impossible for inaccessible or damaged sites.

3 Methodology

- Data Collection**
 - High-resolution photographs of carved inscriptions and projections from 3D scans of khachkars and tapanakars
 - Manual annotation of symbol masks to serve as ground truth
- Preprocessing Pipeline**
 - Adaptive Thresholding: Adjusts to varying stone surface conditions
 - Morphological Operations: Enhances the structure of carved characters
 - Connected Components Analysis: Identifies candidate regions
- Clustering and Filtering**
 - DBSCAN clustering of components based on spatial and size features
 - Filtering of non-symbol artifacts through size, shape, and texture heuristics
- Recognition**
 - Binary classification model to separate valid glyphs from noise
 - Initial experiments with SVMs, Random Forests, and lightweight CNNs

2 Objective

- Develop a computational framework for detecting and recognizing Armenian stone inscriptions
- Explore classical computer vision and machine learning methods to preprocess, segment, and classify glyphs
- Create preliminary datasets and evaluation pipelines for Armenian epigraphic OCR



4 Results

- Created a custom dataset of Armenian stone inscription images with manual and semi-automatic annotations
- Developed a region proposal pipeline for glyph detection using adaptive thresholding, morphological operations, and clustering
- Detection component now reliably segments individual glyphs across diverse stones and conditions
- Recognition model shows promising results on initial tests, with accurate classification of many common characters

5 Conclusion

Classical computer vision techniques, enhanced by adaptive parameterization, offer a powerful first step toward automating the detection and recognition of Armenian stone inscriptions.

Future Work:

- Expand dataset size with diverse inscription styles and periods
- Integrate deep learning methods for fine-grained classification and error correction
- Explore 3D scanning and photogrammetry for degraded surfaces

References

- [1] Chahan Vidal-Gorène, Aliénor Decours-Perez. Detecting and Deciphering Damaged Medieval Armenian Inscriptions Using YOLO and Vision Transformers
- [2] Hamest Tamrazyan, Gayane Hovhannisyan. Digital Guardianship: Innovative Strategies in Preserving Armenian’s Epigraphic Legacy
- [3] 3DArmenia. <https://tapanakars.3darmenia.com>, <https://khachkars.3darmenia.com/>